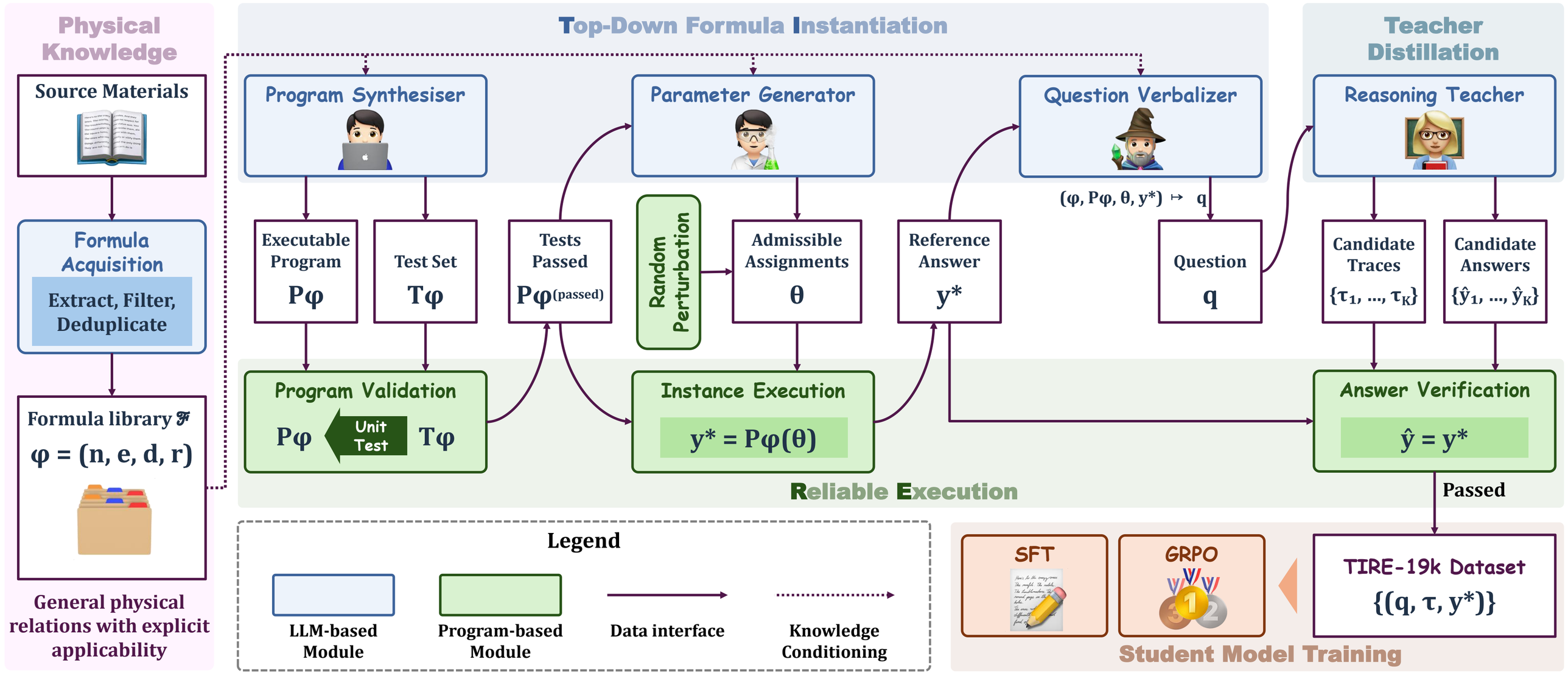

Physical Knowledge
Source Materials
Formula Acquisition
Extract, Filter, Deduplicate
Formula library $\mathcal{F}$
$\varphi = (n, e, d, r)$
General physical relations with explicit applicability
Top-Down Formula Instantiation
Program Synthesiser
Parameter Generator
Question Verbalizer
$(\varphi, P\varphi, \theta, y^*) \mapsto q$
Executable Program $P\varphi$
Test Set $T\varphi$
Tests Passed $P\varphi^{(passed)}$
Random Perturbation
Admissible Assignments θ
Reference Answer y^*
Question q
Program Validation
$P\varphi$ Unit Test $T\varphi$
Instance Execution
$y^* = P\varphi(\theta)$
Reliable Execution
Teacher Distillation
Reasoning Teacher
Candidate Traces $\{\tau_1, ..., \tau_K\}$
Candidate Answers $\{\hat{y}_1, ..., \hat{y}_K\}$
Answer Verification
$\hat{y} = y^*$
Passed
Legend
LLM-based Module
Program-based Module
Data interface
Knowledge Conditioning
SFT
GRPO
TIRE-19k Dataset
$\{(q, \tau, y^*)\}$
Student Model Training